

Nanocomposites for Delivering Agrochemicals: A Comprehensive Review

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ABSTRACT: Excessive application of fertilizers negatively affects soil health, causes low nutrient utilization efficiency in plants, and leads to environmental pollution. The application of controlled-release fertilizer is gaining momentum to overcome this crisis. Engineered nanocomposites (ENCs) have shown tremendous promise for need-based delivery of agrochemicals (macro- and micronutrients, pesticides, and other agrochemicals). This review provides comprehensive coverage of synthesis of nanocomposites, their physical–chemical characterization, and techniques to achieve sustained release and targeted delivery to the crops, emphasizing their beneficial role in plant production and protection. Related aspects like feasibility of the application, commercialization of the nanoformulations, and biosafety concerns are also highlighted. This will be helpful to develop a critical understanding of the current state of the art in the controlled release of agrochemicals through nanocomposites. The pressing issues like scale up production, cost analyses, field-based trials, and environmental safety concerns should be given greater attention in future studies.

KEYWORDS: engineered nanocomposites, controlled release, targeted delivery, agrochemicals, commercialization, environmental safety

■ INTRODUCTION

The world population is expected to be 9.6 billion by 2050, and it has been predicted that the global grain production needs to be increased by 70% to meet food demands for the rapidly increasing world population.^{1,2} However, on the other hand, food supply is likely to face a decline due to a reduction in agricultural yields, limited land availability, and other factors. Although in the past few decades application of fertilizers had been instrumental for increased agricultural productivity, recent studies regarding the long-term effects of such agrochemicals confirm a decline in soil health.³ Yet, to ensure the increase in productivity of crops, farmers often use fertilizers, pesticides, and herbicides in excessive amounts. Indiscriminate usage of such agrochemicals should be avoided owing to their carcinogenic and mutagenic properties, and hazardous effects in human health and environment.^{4,5} In addition to this, conventional agrochemicals have several disadvantages; application of these fertilizers in the soil causes an initial spike in concentration levels, but with time, it rapidly declines to below effective level, due to degradation, leaching, or volatilization of that compound. This phenomenon leads to repeated application of the conventional fertilizers, which eventually results in a decline in soil quality and eutrophication of nearby water bodies.

For soil nutrient supplementation, a huge amount of fertilizers, in the form of ammonium salts, urea, nitrate, or phosphate compounds, are applied in the fields leading to extremely high local concentrations that negatively affect the crops.⁶ Most of the applied urea is also lost via volatilization, leaching, etc., and accumulation of NH_4^+ causes an increase in soil pH. Nitrogen loss from surface soil by volatilization can reach up to 40–70%.⁷ Phosphate fertilizers are also rendered unavailable to the plants as they tend to form insoluble inorganic

compounds, mainly Fe- and Al-based oxides, which decrease the soil phosphate levels up to 0.01 to 1.00 ppm, and 80–90% of applied phosphate fertilizers are lost to the environment. In addition to this, the presence of excess ammonia and sodium can inhibit the uptake of potassium, which leads to very low (18–20%) utilization efficiency of potassium fertilizers.^{8–10} The deficiency of micronutrients also affects crop yield. Major micronutrients like iron are mainly unavailable to plants in calcareous soil, primarily due to the low solubility of the oxidized ferric form in aerobic environment.¹¹ Zinc and manganese deficiency is also common in neutral and alkaline (pH) soil and in calcium rich soil.¹²

Hence, it is necessary to develop alternative fertilizers through the exploitation of nanotechnology and nanomaterials.¹³ Recently, nanotechnology has paved a new era of controlled release systems. Controlled release can be defined as permeation regulated transfer of active ingredients from a modified reservoir to a specific target region and, at the same time, concentration level of the active ingredient is maintained at a predetermined level for a prolonged time period.¹⁴ Several nanocomposites have been designed for slow delivery of agrochemicals, and since they possess high durability, extended shelf life makes them quite suitable for their application in agricultural field conditions.^{15,16} In the context of sustainable agriculture, the application of controlled release systems is regarded to be one of

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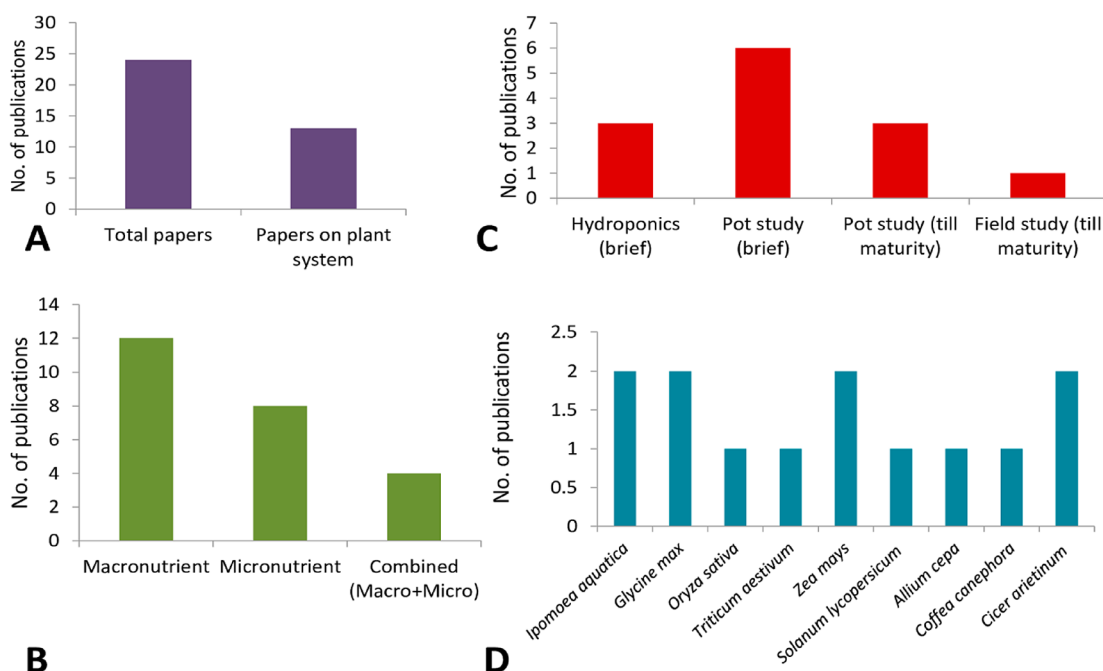


Figure 1. Graphical representation of reports on nanocomposites. (A) Total number of publications on nanocomposite and its simultaneous applications in plant systems. (B) Number of nanocomposites for macro- and micronutrient delivery. (C) Experiments in hydroponic system, pot, and field studies. (D) Experiments in different plant systems.

the promising approaches to increase nutrient availability to the plants.¹⁷ Several macronutrients, micronutrients, nutrient loaded nanoparticles, and plant growth enhancing nanomaterials have been engineered to increase the yield of crops.¹⁸ Nutrients can be easily released from these ENC's (engineered nanocomposites) in the form of soluble ions after they are dissolved in the solution. Nanofertilizers are also considered to be more effective than the traditional fertilizers due to their remarkable impact on crop nutritional qualities and stress tolerance in plants.¹⁹

The majority of reports regarding nanocomposites are limited to synthesis, characterization, and release kinetics studies in soil and water. Only a handful of studies are available that reported the application of nanocomposites on plants (Figure 1).

This review is aimed to provide an overview of different nanocomposites designed specifically as either controlled release fertilizer or delivery agents of different agrochemicals. At the same time, their applications in different crop plants are also highlighted (covering the last five years). Reports regarding the application of the nanocomposites in field or pot experiments confirming the efficacy of the slow release systems in increasing the overall plant growth are also included in this discussion.

ENGINEERED NANOCOMPOSITE FOR CONTROLLED RELEASE OF NUTRIENTS

In recent decades, several smart nanocomposites are being designed as carriers for fertilizers. The nutrient elements can be administered in different ways:

- direct delivery of nutrients in nanoscale dimension as particles or emulsion;
- encapsulated nanofertilizers designed specifically for controlled release; controlled delivery of nutrient can be achieved by different modes,

- slow release systems, active ingredient released from the capsule over a longer time,
 - quick release, active ingredient released upon contact with the target surface,
 - conditional release systems, based on pH/temperature/moisture, magnetic/ultrasonic pulse based release
- (c) nutrient delivery via complex nanocapsule, incorporated in the matrix of an organic polymer of biological or chemical origin, which serves as a carrier

Nanoencapsulated products have recently been used in the delivery of agrochemicals since they not only ensure slow release but also provide protection to the delivery cargo at its core. Here, the delivery cargo is coated with a hydrophilic shell, which actively absorbs water resulting in swelling and the subsequent dissolution and release of the cargo by diffusion.²⁰

NANOENCAPSULATION TECHNIQUES

For the construction of efficient nanoencapsulated techniques, several factors, like particle size, distribution, solubility, shape, surface area, encapsulation mechanism, and release mechanism, should be considered critically. Some major nanoencapsulation techniques that are frequently used are emulsification, coacervation, inclusion complexation, solvent displacement or nanoprecipitation, and supercritical fluid techniques. Nanocapsules of 10–1000 nm size range can be produced by these techniques.

- Emulsification. Nanoemulsion are thermodynamically stable isotropic systems where two immiscible liquids are mixed to form a single continuous phase with the help of an emulsifying agent (surfactant or cosurfactant). Nanoemulsions can be used in liquid form or can be converted to a powder form by spray-drying or freeze-drying techniques after emulsification. To construct nanoemulsions, high-energy emulsification methods like

Table 1. Advantages and Disadvantages of Different Nanocarriers Used for Designing Nanocomposites

carrier system	advantage	disadvantage	citation
silica	easy to synthesize and to maintain desired size shape and structure		35, 36
chitosan	- boost plant tolerance against stress - biodegradable - antimicrobial activity, ecofriendly - sparingly soluble in aqueous solution		37, 38
solid lipid NPs	- entrapment and controlled release of lipophilic delivery cargos - ecofriendly - no need for organic solvent during synthesis	- low-loading efficiency - leakage after long storage	26
layered double hydroxides	can cross plant cell wall barrier owing to its positive charge	- breaks down under acidic conditions - addition of moisture and carbon dioxide	39, 40
carbon nanotubes	- beneficial for plant - enhance water absorption - easily penetrate plant cell	phytotoxic in some reports	41
nanoclay	- adsorb water and can retain for long time - suitable for plant need based release - ecofriendly and biocompatible	further modification and improvement of synthesis procedure aiming for large scale production required	42
nanozeolite	- helps to promote plant growth - promotion of soil rhizobacteria growth to improve plant and soil health - ecofriendly	high synthesis cost	43

hot or cold homogenization, high pressure homogenization, solvent emulsification–diffusion, solvent emulsification–evaporation, melting dispersion, ultrasonication, solvent injection, and double emulsification are used. These methods provide the highest amount of energy in the shortest time, which provides a homogeneous flow to form the smallest droplet sizes.²¹ Emulsification is a highly effective and simplistic technique to construct an agrochemical delivery system. Nanoemulsions not only ensure higher bioavailability of the lipophilic active ingredients encapsulated within but also possess high stability to particle aggregation and gravitational separation.^{22,23}

- (b) Coacervation is also known as a phase separation technique used for encapsulation of hydrophilic molecules and is a highly efficient but costly technique.²⁴ Coacervation is a process by which a homogeneous solution of charged macromolecules goes through liquid–liquid phase separation and forms a polymer-rich, dense phase. Different natural hydrophilic biopolymers like sodium alginate, gelatin, and chitosan are used for the development of biopolymeric nanoparticle systems by this method. The sizes of nanocapsules produced by the coacervation technique range from 100 to 600 nm but are chiefly dependent upon the drying techniques like vacuum drying and freeze drying.²⁴ The coacervation technique is considered to be very promising due to its high payload delivery capacity (up to 99%) and the possibility of developing controlled release systems based on temperature, pH, mechanical stress, etc.²⁵
- (c) Inclusion complexes are also called occlusion compounds where a nonpolar compound (guest) is enclosed inside the nonpolar cavity of another molecule or group of molecules forming an open lattice or cage like crystal structure (host). The interaction between guest and host is maintained by electrostatic force, hydrogen bonding,

van der Waals force, or entropy driven hydrophobic interaction. Although this mechanism of nanoencapsulation ensures very high stability, it is not very useful for designing nanocomposites for agrochemical delivery.

- (d) Nanoprecipitation or solvent displacement technique is the most commonly used and efficient technique for polymeric nanoparticle (sizes below 100 nm of spherical shape) synthesis with encapsulation efficiency of about 75–96%, sustained release, and increased absorption.²⁴ This process involves the emulsification of polymers in an aqueous phase followed by dispersion in a volatile solvent (chloroform, ethyl acetate, etc.).
- (e) Nanoliposomes are nanosized liposomes (spherical amphiphilic structures made from lipids) used for the delivery of biologically active agents or agrochemicals. In a liposome, hydrophilic heads are oriented outside forming the outer shell and hydrophobic tails are arranged inward forming the central core. Liposome mediated delivery is quite lucrative because it can be modified to deliver both hydrophobic and hydrophilic bioactive agents. Frequently employed methods for liposome construction are thin film hydration, detergent removal, solvent injection, and reverse phase.²⁶

■ POLYMER BASED NANOCOMPOSITES AND THEIR SYNTHESIS

Several polymer-based nanocomposites are also being used to design nanocomposites for the delivery of agrochemicals. Sol–gel technique is a well-known bottom-up synthesis technique for nanoparticles. It involves several stages like hydrolysis and polycondensation, gelation, aging, drying densification, and crystallization. Silica-based hybrid nanomaterials and zeolites are mostly prepared by sol–gel technique.²⁷

Polymer–silicate nanocomposites can be synthesized by different techniques: exfoliation–adsorption, in situ intercalative polymerization, melt intercalation, template synthesis, and

Table 2. Synthesis and Characterization of Nanocomposite Fertilizer

nanocomposites	synthesis technique	ref
urea–aminopropyltrimethoxysilane (APTMS)–zeolite	APTMS surface modified zeolite with urea as slow release fertilizer	45
nanozeolite loaded with macronutrients (N, P, K, Mg, Ca, S) and micronutrients (Fe, Zn, Cu)	nanozeolite by coprecipitation followed by impregnation of macronutrients (N, P, K, Ca, Mg, S) and micronutrients (Fe, Zn, Cu).	46
surface modified zeolite loaded with nitrogen	hexadecyltrimethyl ammonium bromide modified zeolite with nitrogen	47
zeolite loaded with phosphorus and potassium fertilizer	phosphorus and potassium loaded onto the surfactant modified zeolite	48
hydrogel based nanocomposite for urea delivery	chemical polymerization of acrylamide monomer to form hydrogel	49
clay loaded hydrogel based nanocomposite for urea and boron delivery	nanocomposite comprising PAAm (polyacrylamide) and biodegradable polysaccharide CMC (carboxymethyl cellulose) by chemical polymerization	50
hydrogel based nanocomposite for NPK delivery	hydrogel formulation with APS and AAm and cross-linkers AA and MBA, followed by hydrogel adherence with NPK	51
NPK fertilizer encapsulated by carboxymethyl cellulose	hydrogel formulation with poly(vinyl propylene) (PVP) and silica nanoparticles and incorporation of NPK fertilizer	53
apatite nanoparticles as a phosphorus fertilizer	carboxymethyl cellulose stabilized apatite nanoparticles, followed by addition of PO_4^{3-} solution	55
liposome mediated delivery of magnesium and iron	liposomes formulation followed by incorporation by Mg and Fe	61
urea-modified hydroxyapatite (HA) nanoparticle encapsulated in wood	HA nanoparticles by chemical precipitation with calcium hydroxide and orthophosphoric acid; <i>Gliricidia sepium</i> soft wood for encapsulation of HA nanoparticles	56
hydrogel based delivery of iron fertilizer	clay, carra, and MBA to formulate hydrogel nanocomposite	62
ferrous foliar fertilizer	pH-controlled-release ferrous foliar fertilizer (PCRFFF) composed of ferrous sulfate, microcrystalline cellulose, and attapulgite	63
iron fertilizer	temperature controlled release and collectable iron fertilizer (TCIF) by synthesis of PFF particles with iron chloride precursors and PAL and formation of PFF@ASO	64
zinc–boron nanocomposite with chitosan nanoparticle	chitosan nanoparticle emulsion with TPP ionic gelatin; Zn/B nanofertilizer formulation in the chitosan nanoparticle suspension	66
Cu–Zn/CNF nanocomposite	impregnation of Cu and Zn salts on activated carbon fibers; CVD on Cu–Zn/ACF to form CNF supported Cu–Zn nanocomposite	68
nanocomposite with micronutrient nanoparticles ZnO , B_2O_3 , and CuO	microbit formulation by mixing of metal oxide nanoparticles zinc oxide, copper oxide, and boron oxide	69
iron/NPK formulation from superabsorbent nanocomposite	multinutrient fertilizer preparation by ferric chelate, urea, ammonium dihydrogen phosphate, and potassium dihydrogen phosphate with hydrogel solution	60
nanocomposite for plant disease treatment	preparation of MWCNT-g-PCA–EP from MWCNT-g-PCA hybrid material; incorporation of mancozeb and zineb pesticides onto CNT-g-PCA	75

sol–gel technique. Among all the methods, the melt intercalation synthesis method is the most simple, economical, and ecofriendly procedure.²⁸ Hence it is considered as a widely accepted synthesis protocol for producing polymer–silicate nanocomposite.²⁸ Generally, in this process, the polymer is melted and combined with desired proportions of intercalated clay by using an extruder. Based on the compatibility between the silica layers and the polymer, molten or intercalated nanocomposite can be obtained. Since this entire process is carried out in an inert gas, organic solvent is not required.

Polymer nanocomposites using carbon nanotubes (CNTs) as filler were first reported in 1994.²⁹ However, the dispersion of carbon tubes in polymeric matrices is quite difficult. Research on the functionalization of CNTs is being performed to improve the dispersion of CNTs in a polymer matrix. CNT functionalization mainly comprises 2 basic methods, covalent or noncovalent modifications.^{30–32}

■ CARRIER SYSTEMS IN NANOCOMPOSITES

In addition to silica and layered double hydroxides, natural biopolymers like alginate, starch, cellulose, cyclodextrin, dextran, guar gum, pectin, and chitosan are commonly used as carriers to encapsulate or absorb active molecules to develop efficient agrochemical delivery systems (Table 1). These compounds are widely available in nature, and the synthesis of such nanocomposites is also cost-effective.³³ Nanoclays and zeolite are also widely used as carriers owing to their crystalline honeycomb like layered structure, which provides a greater surface area and can be easily loaded with agrochemicals.³⁴ Carbon nanotubes (CNTs) or carbon nanofibers (CNFs) are also used as carriers

for nanofertilizers because they can easily penetrate the plant cell for delivery of plant nutrients.

Owing to the unique properties of nanocomposites such as slow release and targeted delivery of agrochemicals, use of nanocomposites can be hugely beneficial compared to bulk chemical fertilizers and pesticides. Synthesis processes of some nanocomposites designed for controlled release of agrochemicals are mentioned in Table 2, and the role of nanocomposites in agriculture has been critically evaluated in the following sections.

■ NANOCOMPOSITES AS FERTILIZERS

Among macronutrients, nitrogen, phosphorus, and potassium are vital for plant growth and maintenance. Nitrogen and phosphorus have major roles in the structural development of plants, for the synthesis of nucleic acids, protein, etc., and potassium also plays a major role in maintaining osmotic potential and ion balance.⁴⁴ Several nanocomposites have been designed to ensure the effective delivery of these macronutrients in agricultural fields.

Macronutrient Delivery Systems. Slow release of fertilizers from nanocomposites depends on the stability of coatings, which plays a major role in reducing the dissolution rate of fertilizers. According to several reports, zeolite can be used as a carrier for nutrients. Amino-propyl-trimethoxy-silane (APTMS) surface modified zeolite has been used as a carrier to design nanofertilizer for urea delivery.⁴⁵ Another unique nanocomposite system has been designed using nanozeolite as a carrier for macronutrients (N, P, K, Ca, Mg, and S) and micronutrients (Fe, Zn, and Cu).⁴⁶ Nanozeolite is synthesized by a simple coprecipitation method, followed by impregnation

of the macro- and micronutrient salts ($\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{Ca}_3(\text{PO}_4)_2$, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, KCl , NaNO_3 , and $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$). Nutrient release studies in both soil and water confirmed that nanozeolite can maintain a steady supply of nutrients for a longer duration. Due to the porous nature of zeolite, water absorbance, swelling ratio, equilibrium water content, salt index, and water retention studies showed satisfactory results. Two other zeolite based nanocomposites have been designed for NPK delivery.^{47,48} The first nanocomposite was designed solely as a nitrogen delivery system. Here, hexadecyltrimethylammonium bromide (HDTMABr) has been used as a surfactant for the modification of zeolite, which was then loaded with nitrogen. A pot experiment with *Ipomoea aquatica* for 30 days was carried out to evaluate the efficiency of the nanocomposite. The results showed that nitrogen uptake by the plants treated with nanocomposite was higher compared to those treated with conventional urea. Another zeolite based nanocomposite was designed for the delivery of phosphorus and potassium. Pot experiments showed increased uptake of phosphorus and potassium by *Ipomoea aquatica* after they were grown for 30 days.

Several slow-release fertilizers encapsulated by superabsorbent capsules have been reported. The superabsorbent hydrogels have a dual advantage as they help to increase soil fertility and reduce the need for water supply by irrigation as well. Recently, hydrogel polymer has emerged as a lucrative alternative for creating fertilizer delivery systems. Hydrogel polymers have the innate properties of water absorption and biodegradability and can also act as slow delivery agents. A novel nanocomposite system for delivery of urea, based on a hydrogel that is chiefly composed of polyacrylamide, along with biodegradable polysaccharide methylcellulose and calcium montmorillonite acting as modifiers, has also been designed.⁴⁹ Similarly, this clay loaded hydrogel based nanocomposite has been also used for effective delivery of urea and boron.⁵⁰ Another hydrogel-based NPK fertilizer delivery system developed by polymerization of sodium alginate, acrylic acid, acrylamide, and clinoptilolite zeolite.⁵¹ The nanocomposite also showed enhanced water absorption and retention properties, and fertilizer release percentage in soil was around 75% on the 30th day. A unique NPK loaded superabsorbent nanocomposite chiefly based on maize bran, acrylic acid and acrylamide, and montmorillonite has been also reported.⁵² The water absorbance capacity of the nanocomposite was dependent on montmorillonite content and N,N' -methylene bis(acrylamide) content. A novel slow release fertilizer has been prepared by in situ polymerization of sulfonated carboxymethyl cellulose with acrylic acid along with poly(vinyl pyrrolidone), silica nanoparticles, and NPK fertilizers (from urea, ammonium dihydrogen phosphate, and potassium dihydrogen phosphate).⁵³ The release kinetics study confirms the slow release of fertilizers, and the nanoformulations also exhibited pH and salt sensitivity and excellent water retention properties. Kaolinite, a layered silicate clay mineral, has been used recently for the synthesis of urea nanocomposite.⁵⁴ The nanocomposite is encapsulated by a gum-arabic matrix to ensure a slow release of urea.

Hydroxyapatite nanoparticles are rich sources of phosphorus, which is reported to be remarkably efficient in increasing the yield of soybean (*Glycine max*) plants.⁵⁵ Compared to the commercial phosphate fertilizers (Triple superphosphate, $\text{Ca}(\text{H}_2\text{PO}_4)_2$), the growth rate and height of the plants were significantly higher in hydroxyapatite nanoparticle treated plants. Utilization of this nanofertilizer can also reduce the

environmental hazards of eutrophication. Hydroxyapatite in its nanoform has much weaker interaction with soil components than the charged ions of PO_4^{3-} , HPO_4^{2-} , H_2PO_4^- , or Ca^{2+} . Therefore, it can deliver a greater amount of phosphorus and calcium sources to the plant, which enhances crop yield.

Urea coated hydroxyapatite nanoparticles are capable of slow release of nitrogen in the soil.⁵⁶ Hydroxyapatite [$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$] serves as a rich phosphorus source, and due to its high surface area, it can also bind to a large amount of urea. The nanofertilizers showed an initial burst and then slow release of nitrogen until the 60th day of application. This sustained release of nitrogen is quite advantageous compared to commercial fertilizers because they only release the nitrogen heavily during the early phase and then release low and nonuniform quantities only until the 30th day. Field trial for rice cultivation showed that the use of these monohybrids not only enhanced plant growth and crop yield (0.6 tons/hectare increase compared to the yield obtained by using recommended urea amount) but also contributed to a 50% reduction in urea use.

Recently chitosan has also evolved as a promising polymer for fertilizer delivery. Foliar application of nanofertilizers is considered another alternative approach to nutrient delivery. Here, nanoparticles can enter through stomatal opening and bases of trichomes and finally gets translocated to various tissues of the plant by the vascular system.⁵⁷ Nanochitosan has been used as a NPK delivery system by foliar uptake in wheat plants.⁵⁸ Higher harvest index, crop index, and mobilization index proved the efficiency of the nanofertilizers over the conventional bulk NPK fertilizers in increasing crop production. The uptake of nutrients was confirmed from TEM images of leaf phloem tissue, which showed internalization of nano-NPK materials (26.2–30.6 nm in diameter).

Several multinutrient-loaded nanocomposites have also been designed. Chitosan and graphene oxide coated potassium nitrate nanocomposite has been designed.⁵⁹ The fertilizer beads were coated with chitosan–graphene oxide film by a simple dipping method. The unique delivery system for NPK and iron, which is prepared using a maize bran–poly(acrylic acid-co-acrylamide)/montmorillonite superabsorbent nanocomposite was also reported.⁶⁰ The nanocomposite possessed high thermal stability and water absorption capacity. In addition to the slow release of fertilizer, the nanocomposite showed remarkable pH dependent swelling reversibility and high salt resistance properties. A unique nanocarrier that was basically a 100 nm liposome loaded with magnesium and iron showed that the liposome was translocated through the leaf to other regions and subsequently the nutrient was released in the nucleus or cytoplasm of tomato. The nutrient deficient tomato plants upon the treatment with liposomal iron and magnesium formulation showed normal turgor pressure in the leaves and recovered from chlorosis and epinasty.⁶¹

Micronutrient Delivery Systems. Micronutrients like iron, zinc, magnesium, molybdenum, copper, and boron are required in trace amounts for proper growth and development of the plants. There are several reports where it is shown that the application of these micronutrients in nanoform can enhance plant growth.

Iron. Among several essential plant micronutrients, iron deficiency is a common problem, even though iron remains in the soil in a very high amount, as most of the iron remains unavailable for plant uptake in neutral and alkaline soil conditions. Hence, to solve this problem, several slow release iron fertilizers have been designed.

Table 3. Nanocomposite Fertilizers and Their Effects on Plant Growth

nanocomposite	treatment concentration	plant species	growth condition and duration	effects on plants	citation
nanozoelite modified with HDTMABr loaded with N	250 mg/kg	<i>Ipomoea aquatica</i>	pot study (30 days)	plant growth ↑, biomass ↑, nitrogen uptake and concentration ↑	47
nanozoelite modified with HDTMABr loaded with P and K	15 mg/pot	<i>Ipomoea aquatica</i>	pot study (30 days)	plant growth ↑, biomass ↑, phosphorus phytoavailability 2.3-fold ↑, potassium phytoavailability 1.1-fold ↑	48
hydroxyapatite NPs Diameter: 15.8 ± 7.8 nm	21.6 mg L ⁻¹	<i>Glycine max</i>	pot study (until maturity)	plant growth rate 32.6% ↑, biomass 18.2% ↑, yield 20.4% ↑	55
urea coated hydroxyapatite nanoparticles; diameter 15–20 nm, length 100–200 nm (rod like)	50 kg h ⁻¹	<i>Oryza sativa</i>	field conditions (until maturity)	yield ↑ (0.6 t h ⁻¹ higher than conventional urea)	56
chitosan–poly(methacrylic acid) NPs loaded with NPK; diameter, 26.2–30.6 nm	10%, 25%, and 100% nano-NPK suspension	<i>Triticum aestivum</i>	pot study (till maturity)	growth ↑, yield 10% ↑, agronomic parameters ↑	58
liposome mediated delivery of Mg and Fe	20–30 μL of 0.2% formulation applied every 48 h (foliar spray)	<i>Solanum lycopersicum var. ceriseforme</i>	hydroponics (8 weeks, until 8–12 leaf stage)	growth ↑, leaf number ↑ (under iron deficient conditions), uptake efficiency of the liposome 33% (after 72 h treatment)	61
pH controlled release of ferrous foliar fertilizer	5 g L ⁻¹ of 5 mL aqueous solution (pH 6) (foliar spray)	<i>Zea mays</i>	greenhouse pot experiment	growth ↑, chlorophyll content ↑ (compared to traditional fertilizer)	63
temperature responsive controlled release iron fertilizer	1.5 g kg ⁻¹	<i>Zea mays</i>	greenhouse pot experiment (12 days)	growth ↑, biomass ↑, chlorophyll ↑, iron ↑, maximum efficacy at 35 °C	64
nanocomposites with MWCNTs and ZnO NPs	15 μg mL ⁻¹	<i>Allium cepa</i>	hydroponics (10 days)	germination (%) ↑, growth ↑, cell division ↑	65
Zn–B nanocomposite with chitosan nanoparticle	20–30 ppm, used 3 times within 20 days (foliar spray)	<i>Coffea canephora</i>	greenhouse pot experiment (120 days)	growth ↑, photosynthetic efficiency ↑, N, P, Zn uptake ↑	66
Cu–CNF nanocomposite	10–500 μg mL ⁻¹	<i>Cicer arietinum</i>	hydroponics (25 days)	water uptake ↑, germination rate ↑, growth ↑, chlorophyll and protein contents ↑	67
Cu–Zn/CNF nanocomposite based on poly(vinyl alcohol) and starch blend	0.25–4 g kg ⁻¹	<i>Cicer arietinum</i>	pot study (35 days)	growth ↑, biomass ↑, relative water levels ↓, chlorophyll and protein contents ↑, ROS levels ↓	68
nanocomposite with micronutrient nanoparticles ZnO, B ₂ O ₃ , and CuO	Zn, Cu, B 2.8, 1.3, 0.6 mg kg ⁻¹ , respectively (foliar spray) after 21 days	<i>Glycine max</i>	greenhouse pot experiment draft (field moisture capacity 50%) from 35th day (until maturity)	growth ↑, grain yield ↑, 36% N ↑, 28% K ↑, 19% Zn ↑, 1080% B ↑, 74% Cu ↑, 954% grain nutrient content ↑, N ↑ 35%, K ↑ 32%, Zn ↑ 68%, B ↑ 56%, Cu ↑ 13%, draft stress ↓	69

A hydrogel based nanocomposite has been deployed for controlled release of iron fertilizer sequestrene Fe 138. The nanocomposite is based on κ -carrageenan and composed of sodium montmorillonite, along with acrylamide and methylene bis(acrylamide) as a monomer and cross-linker, respectively.⁶²

pH-controlled release of ferrous foliar fertilizer has been designed recently. The nanocomposite was primarily made with microcrystalline cellulose and attapulgite. To construct the nanocomposite system, first, the cellulose was oxidized to carboxyl cellulose, and the presence of $-\text{COOH}$ groups facilitated the attachment of Fe^{2+} ions, thus forming the Fe-coated microfibers. The microfibers were in turn coated with attapulgite via hydrogen bonds. The pH-dependent release of ferrous fertilizer is mainly achieved from this attapulgite coating because under low pH the attapulgite loosens thus favoring iron release. The effectiveness of this nanocomposite as a foliar spray has also been proven via a pot study on maize. The application of nanocomposite greatly enhanced the utilization efficiency of iron, which is reflected in enhanced plant growth and chlorophyll pigment contents.⁶³

Temperature responsive controlled release iron fertilizer has been designed. This unique nanocomposite is a core-shell structure, where palygorskite (environmentally friendly, cost-effective, and stable fibrous nanoclay, acting as the carrier), Fe_3O_4 , and ferrous ammonium sulfate hexahydrate formed the core and the shell is composed of amino silicone oil and ethylene oxide/propylene oxide block copolymer (also known as F-127, a thermosensitive polymer used to modify the carrier, palygorskite). To achieve the release of Fe^{2+} from the core of the nanocomposite, thermal treatment at 100°C was carried out to create micropores on the amino silicone oil shell, and F-127 was then further used to mediate the temperature based release of iron. Since F-127 is temperature sensitive, it stays in a liquid state within the temperature range $25\text{--}35^\circ\text{C}$, and at 15 and 45°C , it remains in the gel state. In the liquid state, the micropores of amino silicone oil shells are exposed, which facilitates the release of iron whereas in the gel state, a higher number of pores remain covered resulting in the release of iron in a lower amount. The amount of iron released with respect to four different temperatures follows the pattern $35^\circ\text{C} > 25^\circ\text{C} > 45^\circ\text{C} > 15^\circ\text{C}$. The pot experiments with maize seedlings confirmed the effectiveness of the nanocomposite as an iron fertilizer. Highest plant growth and chlorophyll and iron content have been reported in the seedlings that were grown at 35°C .⁶⁴

Zinc. The supply of zinc fertilizer in nanoform can enhance the bioavailability of the micronutrient to the plants. Synergistic effects of multiwalled carbon nanotubes (MWCNTs) and ZnO NPs as a growth enhancer of *Allium cepa* has been recently reported.⁶⁵ Within the nanocomposite, MWCNTs acted as perfect carrier systems by ensuring the easy penetration and delivery of Zn in the plant cell, and at the same time, it played a role in the prevention of the aggregation of ZnO NPs. It is also reported that treatment with $15\text{ }\mu\text{g/mL}$ ZnO/MWCNTs showed the highest germination percentage of onion seeds.

A zinc-boron nanofertilizer, engineered by incorporating ZnSO_4 and H_3BO_3 on a chitosan nanoparticle emulsion, has been tested on coffee seedlings in greenhouse conditions.⁶⁶ Five different doses (0, 10, 20, 30, and 40 ppm) of the nanofertilizer were used as a foliar spray. Among those treatment doses, $20\text{--}30\text{ ppm}$ showed maximum improvement in growth, nutrient contents, and other photosynthetic parameters.

Copper. Copper is also considered to be essential for plant growth owing to its role in RNA synthesis, photosystem

functions, and plant cell metabolism and signaling. Effective delivery of copper micronutrient in *Cicer arietinum* seedlings with the help of copper nanoparticles grown in carbon nanofibers is a recent achievement. The copper-carbon nanofiber (Cu-CNF) nanocomposite was synthesized from activated carbon microfibers. At first, the microfibers were impregnated with $0.4\text{ M Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ salts in the presence of SDS acting as a surfactant. After calcination and reduction, Cu-activated carbon fiber (ACF) samples were produced, and finally, the CNFs were grown on the ACFs by the process of chemical vapor deposition. Finally, the effect of this copper nanocomposite was monitored in germinating *Cicer arietinum* seedlings. The CNFs acted as an efficient carrier system for copper and also enhanced the water uptake by the plants. Results showed that water uptake capacity, rate of germination, plant growth, and chlorophyll and protein contents of the plants were significantly enhanced.⁶⁷ Another nanocomposite system produced by dispersion of Cu-Zn/CNFs in a poly(vinyl alcohol) and starch blend has been tested. The encapsulation of the micronutrient within biodegradable polymer helped to develop slow release of the micronutrient. The results showed that treatment with the nanocomposite (concentration 4 g/kg) produced higher fresh shoot biomass, water content (%), and chlorophyll and protein contents in 30 day old *Cicer arietinum* seedlings. In addition to this, the increased availability of zinc leads to enhanced detoxification of reactive oxygen species (ROS).⁶⁸ Comparatively very few nanocomposite materials have been tested in plant systems, which have been described briefly in Table 3.

■ DELIVERY OF OTHER AGROCHEMICALS (PESTICIDES AND HERBICIDES) VIA NANOCOMPOSITE SYSTEMS

The productivity of crops depends not only on the presence of nutrients but also on other biotic factors. Weeds, animals, pests, and pathogens as the biotic factors can cause $20\text{--}40\%$ crop loss globally, which is huge.^{70,71} Therefore, plant protection is equally important for food security. However, pesticides and herbicides are sometimes harmful and may pose a major ecological threat, as very often they contaminate water bodies leading to major water pollution. Hence, smart delivery systems are now being used for minimizing the usage of herbicides.

Recently receptor based herbicides having a specific binding affinity with nanoherbicide molecules like CNTs are being used to kill the viable and dormant weed propagules. Nano-encapsulated herbicides are being developed to target specific receptors of weed roots.⁷² Biodegradable nanocomposites have been also formulated for the controlled release of herbicide. A nanocomposite based on starch gel and montmorillonite clay has been developed for the controlled release of ametryne.⁷³ Nanoencapsulated herbicides are also designed in a unique way where the release of herbicide molecules is completely regulated based on rainfall and temperature.⁷⁴ During the rainy season, the chances of weed growth are maximum; hence such a nanocomposite system is perfect for rain-fed systems.

A light-responsive controlled herbicide particle (LHCP) was developed using a nanocomposite composed of biochar, attapulgite (ATP), glyphosate (Gly), azobenzene (AZO), and amino silicone oil (ASO). Here azobenzene acted as light motivated stirrer to release glyphosate. The herbicide also showed proper adhesion on weed leaves, which would also reduce the loss of herbicide. Pot experiments with LHCP treated

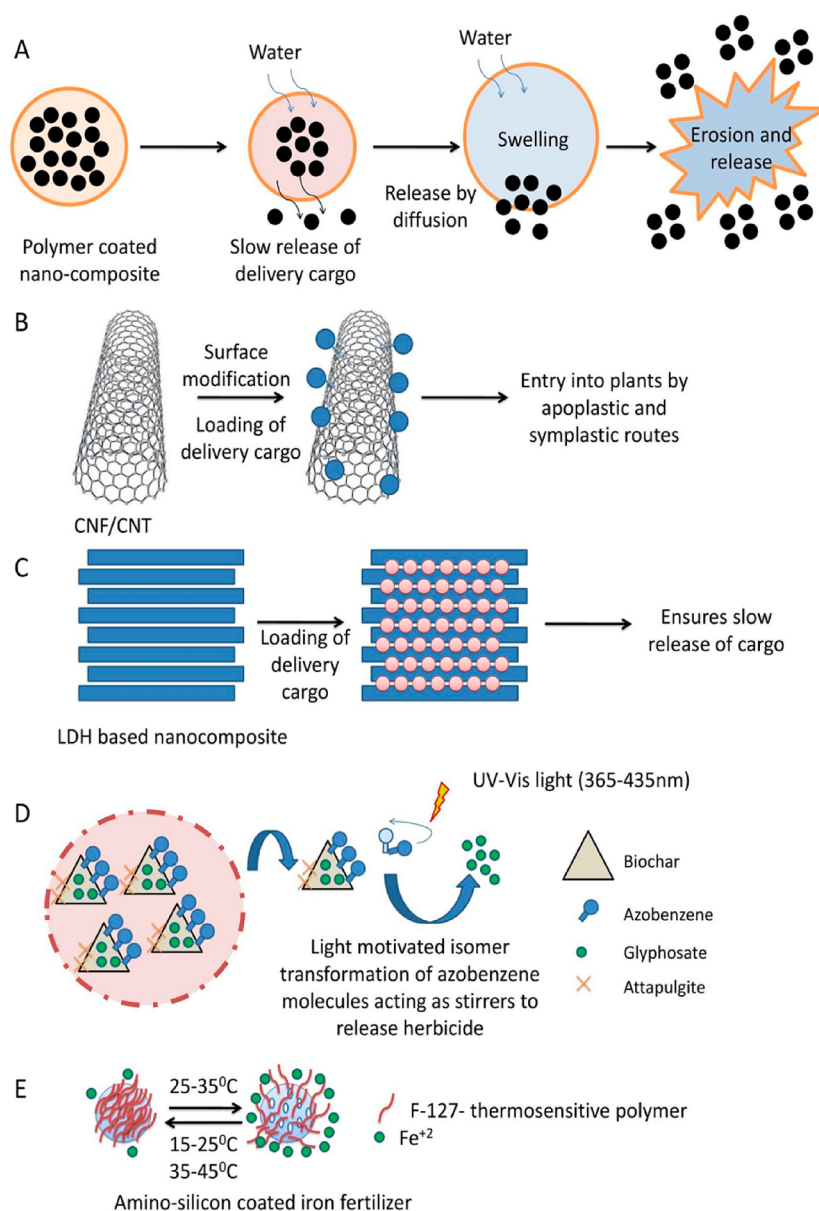


Figure 2. Different nanocomposites used for delivery of agrochemicals (A) Polymer based nanocomposites, which promote slow release of agrochemicals by absorbing water. (B) Surface modified CNT nanocomposites, which can cross plant cell barrier and can be translocated via symplastic and apoplastic pathways. (C) Layered double hydroxide based nanocomposites, which strongly interact with agrochemicals and ensure sustained release of agrochemicals. (D) Light induced release. (E) Temperature induced release of agrochemicals from nanocomposites.

Bermuda weed also confirmed the efficacy of such nanocomposites.⁷⁵

A unique encapsulated pesticide delivery system of mancozeb and zineb pesticides has been also designed.⁷⁶ Multiwalled carbon nanotube-graft-poly(citric acid) (MWCNT-g-PCA) hybrid materials can entrap different water-soluble chemicals and metallic ions. Encapsulation of the pesticide nanoparticles in hyperbranched poly(acetic acid) also increased the effectiveness of mancozeb and zineb compared to the bulk form against the *Alternaria alternata* fungi, which causes early blight disease in many plants. A complex delivery system of acetamiprid has been developed based on alginate and exfoliated montmorillonite (MMT) modified with CTAB, and to ensure the sustained release of the pesticide, acetamiprid was immobilized into the MMT layer by freeze-drying technology.⁷⁷

Different nanocomposites used for delivery of agrochemicals are shown in a schematic diagram (Figure 2).

■ FEASIBILITY OF NANOCOMPOSITE APPLICATIONS IN AGRICULTURE

Studies carried out to test the efficacy of nanocomposites on plant models under field or pot conditions clearly indicate the feasibility of using these nanoformulations in the agricultural sector. Several easy application methods like direct injection through the bark, preplanting seed treatment, or seed priming can significantly enhance the application of nanoformulations in the agricultural sector.⁷⁸ The increased efficiency of nanocomposite fertilizers and their ability to control the precise quantities and conditions under which fertilizers are released could result in a reduction of the volume of fertilizers applied in specific situations, thereby reducing environmental pollution.⁷⁹

Table 4. Nanomaterial-Based Pesticides in the Market

company name	country	product	active ingredient	action mechanism	benefits
Nano Silver Manufacturing Sdn. Bhd.	Malaysia	Nano Green	mixture of metallic oxides with bioceramics	natural antibacterial agent	soil heavy metal ↓, fertility ↑, disease tolerance ↑
Syngenta Co.	USA	PRIMO Maxx	trinexapac-ethyl	decreases production of gibberellic acid	heat, drought, diseases tolerance ↑
Syngenta Co.	USA	Karate ZEON	λ-cyhalotrin	inhibits neural system	control insect pests for barley, wheat, cotton
Agro Nanotechnology Corp.	USA	Nano-Gro	Fe, Co, Al, Mg, Mn, Ni, AgSO ₄	mimics environmental stress to trigger specific plant genes	stress tolerance ↑, growth ↑, yield ↑
HeiQ Materials Co.	Switzerland USA patent (US0294919-A1)	HeiQ-AGS-20	silver–silica nanocomposite	Ag interacts with cellular enzymes and DNA	broad spectrum antibiotic
Nano Sys GmbH	Switzerland	Nano-Argentum 10	silver NP	interacts with cellular functions	antifungal bug repellent

Several cost-effective methods are being used to produce nanoformulations at a much cheaper rate. Indian Farmers Fertilizer Cooperative Limited (IFFCO) is currently taking steadfast measures to ensure the reduction of usage of chemical fertilizers. They have proposed field trials for nanonitrogen, nanozinc (as fertilizers), and nanocopper (as pesticide).

However, most of the works on plant–nanoparticle interactions are focused on nanotoxicity. But it must be also considered that these studies are mostly conducted on plants using very high concentrations of nanomaterials for a very short duration compared to its original life cycle, and in these cases, the studies are conducted in artificial nutrient media. The reports from the results obtained from experiments carried out in nonsoil media can greatly vary with results obtained after studies on agricultural soil because of various physical, chemical, and biochemical interactions. The benefits of nanofertilizers can be truly utilized if more research is conducted to determine the effective doses and time of applications for a definite crop system.

■ ENVIRONMENTAL RISK ASSESSMENT AND SAFETY CONCERNS OF ENGINEERED NANOMATERIALS FOR AGRICULTURAL PURPOSES

Most of the studies on nanofertilizer focused elemental analysis of plants and soil, release kinetics, and leaching, where they failed to show the toxicity level of nanofertilizer.⁸⁰ To check and interpret the toxicity level of micronutrients, it is necessary to analyze the freely dissolved ions in comparison with conventional fertilizer.⁸¹ The information about dissolved ions is vital for environmental risk assessment studies of nanofertilizers. The interaction between nanomaterials and crops includes the transmission of nanoparticle from crops to human beings, cocontamination effects of engineered nanoparticles with other soil pollutants, plant uptake mechanism, and stress response, and the environmental agents that mediate the above process remains poorly studied.⁸² Due to less information about nanomaterial–crop interactions, the safety and quality of food products from nanofertilizer supplied crops are impossible to assess. With the development of new and novel techniques, we can overcome these knowledge gaps. However, to increase the acceptance of nanocomposites in agricultural fields, more detailed studies regarding the safety of these nanocomposites to the environment and public health are required.

■ COMMERCIALIZATION OF NANOCOMPOSITES

It has already been established that nanoformulations are much more effective compared to the bulk forms of agrochemicals. Hence, the global market for nanoformulation production and utilization is increasing rapidly. USA, Brazil, China, India, South Africa, and many European countries have already put significant and strategic investments in developing nanomaterial-based agroproducts. In India, several agrochemical corporations are introducing nanofertilizers.^{83,84} Tropical Agro system India (P) Ltd. has introduced several nanofertilizers under the name TAG NANO (NPK, PhoS, Zinc, Cal, etc.) fertilizers. These are proteino-lacto-gluconate formulations chelated with micronutrients, vitamins, seaweed extracts, humic acids, and probiotics. Another fertilizer company, Nano Green Sciences, Inc., India, has formulated a colloidal nanofertilizer. JU Agri Sciences Pvt. Ltd., Janakpuri, New Delhi, India, and Shan Maw Myae Trading Co., Ltd., have also introduced nanofertilizers under the name Nano Max NPK Fertilizer and Nano Micro Nutrient (Eco Star), respectively. Several leading agrochemical companies like BASF, Bayer Crop Science, Monsanto, and Syngenta have started productions of nanomaterial-based pesticides and herbicides (Table 4).

In summary, this review reports some of the recent achievements in the development of controlled release systems for agrochemicals. It is evident that the application of agrochemicals via control release systems can reduce the overall consumption of commercial fertilizers and pesticides. From an extensive literature survey, it is quite evident that most of the nanoformulations have been designed for the delivery of macronutrients, and there are very few reports based on micronutrient delivery. Further work can focus on developing balanced delivery systems to supply both macro- and micronutrients to plants. Additionally, a large gap exists between “proof-of-concept” studies on nanomaterial- or nanocomposite-based agrochemical delivery systems and commercialization of these technologies. Future research efforts should be directed toward scale up of production, field scale trials, and biosafety assessment to facilitate lab-to-field transition.

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